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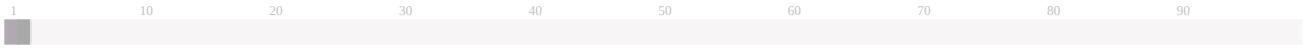
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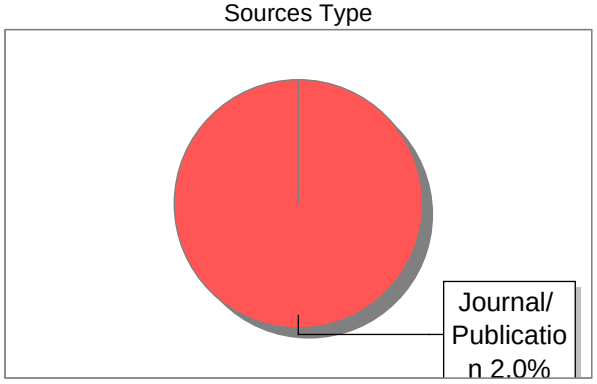
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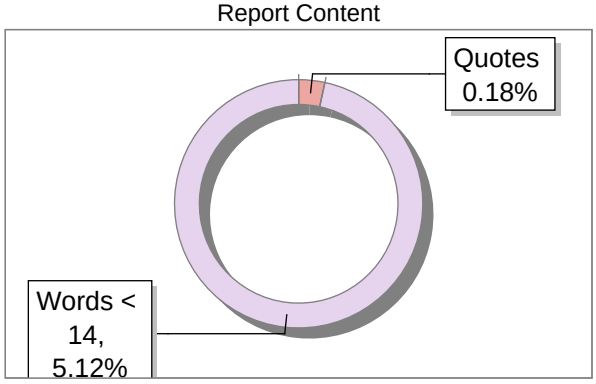
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Agentic Transparency: A Multi-Agent LLM Framework for Regulatory Compliant Credit Risk Explainability

a. Introduction

India's BFSI sector is moving rapidly from branch-led, document-heavy lending toward data-led digital credit assessment. This transition has been supported by digital public infrastructure, wider use of AI, and the growth of NBFC and FinTech lending models. Traditional credit decisions, which were earlier centred on collateral, bureau history, and manual assessment, are now being supplemented by alternative data sources such as UPI activity, digital payment behaviour, telecom patterns, and Account Aggregator records. These data sources allow lenders to assess borrowers who may not have long credit histories, including MSMEs, informal-sector workers, and thin-file retail applicants. As a result, automated underwriting systems are increasingly used to shorten loan assessment timelines and support faster risk classification.

The black-box challenge in credit risk modelling has emerged as a direct consequence of the shift from transparent scorecards to high-performing non-linear models such as XGBoost, LightGBM, and deep neural networks. Although these techniques improve prediction quality and default-risk identification, their decision logic is distributed across many interacting variables and model components. Unlike linear models, where variable weights can be inspected relatively easily, advanced machine learning models do not provide an immediate, intuitive explanation for why a specific borrower is declined. This opacity creates tension between predictive performance and the consumer-protection expectation that lending decisions should be understandable, reviewable, and accountable.

In India, the Reserve Bank of India (RBI) treats algorithmic lending transparency as a regulatory responsibility of the regulated lender rather than merely a technology-design issue. RBI's digital lending framework places accountability for transparent, fair, and ethical lending practices on Regulated Entities (REs), even where digital platforms or lending service providers are involved (Reserve Bank of India, 2022). Similarly, the Fair Practices Code requires lenders to communicate material reasons for loan rejection in writing and in language that the borrower can understand (Reserve Bank of India, 2006). Therefore, a lender cannot rely on a generic statement that a model classified the applicant as risky. The explanation must

identify understandable and actionable financial factors so that the borrower can recognise the basis of the adverse decision and, where possible, improve future credit behaviour.

To meet the need for transparency, researchers and practitioners commonly use post-hoc Explainable AI (XAI) methods, particularly Local Interpretable Model-agnostic Explanations (LIME) (Ribeiro, Singh, & Guestrin, 2016) and SHAPley Additive exPlanations (SHAP) (Lundberg & Lee, 2017). SHAP can estimate the contribution of each feature to a particular model output by drawing on cooperative game-theoretic principles. However, regulatory communication requires more than numerical interpretability. A SHAP output for a rejected credit application may appear as a set of feature names and contribution values, which is useful for analysts but unsuitable for ordinary borrowers. This creates a narrative gap: the model can be explained mathematically, but the explanation still needs to be converted into legally appropriate, plain-language communication at operational scale.

Recent progress in large language models (LLMs) and agentic AI offers a practical route for closing this narrative gap. Existing research indicates that LLMs should not be used as independent credit-risk assessors because they may produce unsupported or inconsistent risk reasoning; however, they can be effective when their role is limited to converting verified model explanations into natural language (Geng et al., 2026). This study therefore proposes a dual-agent design in which the Underwriter Agent performs the predictive and SHAP-based attribution tasks, while a separate Compliance Agent converts the verified attribution outputs into RBI-aligned adverse action narratives. Such a separation of responsibilities supports both model accuracy and regulatory accountability in Indian digital lending.

b. Problem Statement / Research Problem

This research addresses the mismatch between quantitative outputs generated by complex credit-risk models and the borrower-facing transparency expected under Indian financial regulation. As digital lenders adopt non-linear models such as XGBoost and deep neural networks for alternative-credit assessment, they often depend on post-hoc XAI methods such as SHAP to explain model behaviour (Lundberg & Lee, 2017). Yet SHAP primarily produces numerical feature-attribution values. Regulatory expectations under the RBI’s Fair Practices

Code and Digital Lending Guidelines require rejected applicants to receive clear, understandable, and actionable reasons for adverse credit decisions (Reserve Bank of India, 2006; 2022).

In current practice, this translation burden often falls on compliance teams that interpret SHAP charts and manually prepare borrower-facing notices. This manual step weakens the speed and scalability benefits of automated underwriting. Conversely, allowing an LLM to evaluate credit risk independently can introduce hallucinated or unsupported reasoning that may not reflect the actual decision logic of the underlying ML model (Geng, Liu, Li, & Wang, 2026). The resulting research problem is therefore both operational and technological: Indian digital lenders need an automated, privacy-preserving mechanism that can convert verified XAI outputs into compliant regulatory narratives without exposing sensitive borrower data to external systems.

c. Review of the Literature

The growing use of artificial intelligence in financial risk management has required a simultaneous development of Explainable AI (XAI) and privacy-preserving frameworks. A survey of recent literature points to major advances and present constraints in this area.

Capabilities and limits of LLMs in explainability. A central issue in recent literature is whether LLMs can substitute formal mathematical explainers in credit-risk contexts. Geng et al. (2026) examined LLMs as post-hoc explainability tools for credit-risk models and found that they do not reliably reconstruct complex, non-linear feature interactions from raw data alone. Their results indicate that LLM-generated rankings may diverge from model-grounded explanations when the LLM is asked to infer importance directly. However, when SHAP-based reference rankings are supplied through controlled prompts, LLMs are able to preserve the relevant top features and express them in readable language. This supports the view that LLMs are best positioned as narrative interfaces grounded in formal attribution methods rather than as independent mathematical explainers.

Privacy-preserving explainability in distributed finance. Banking data cannot be freely pooled across institutions because of confidentiality, consent, and data-protection concerns. For this reason, federated learning has become relevant to financial AI research. Aljunaid, Almheiri, Dawood, and Khan (2025) show that distributed model training can be combined with explainability methods so that participating institutions can learn from shared patterns while keeping raw customer data within their own environments. Their work is important for this synopsis because it demonstrates that transparency and privacy can be designed together rather than treated as separate requirements.

Class imbalance and heterogeneous financial data. Financial-risk datasets are often uneven because risky outcomes such as default or fraud occur less frequently than normal repayment behaviour. Wu et al. (2026) address this issue in federated financial modelling by proposing a mechanism that gives different importance to data samples according to their information value. This approach helps protect minority-risk signals while maintaining privacy. For the present study, this literature is relevant because it shows that modern credit-risk models are becoming technically sophisticated, making borrower-facing explanation even more necessary.

d. Research gaps

A comprehensive literature review reveals specific academic and operational gaps:

1. **Regulatory translation gap in post-hoc XAI**: Prior work shows that LLMs can generate useful narratives when supplied with SHAP values (Geng et al., 2026), but existing studies generally stop at explanatory summaries. A specific gap remains in mapping model-grounded explanations to the formal borrower-facing language required for adverse action communication under RBI-aligned fair lending practices.
2. **Architectural gap in edge-deployed multi-agent systems:** Existing approaches tend to treat risk scoring, SHAP extraction (Aljunaid et al., 2025), and text generation as separate stages. The literature does not yet provide a complete zero-shot multi-agent architecture in which a predictive underwriter component and a generative compliance component operate together at the edge to produce regulatory documentation immediately after model inference.
3. **Data-sovereign generative compliance gap:** Federated learning addresses privacy during model training (Wu et al., 2026), but borrower-facing explanation may still depend on externally hosted language-model services. The literature does not yet sufficiently demonstrate how an instruction-guided compliance agent can run locally, protect borrower data, and still produce legally reliable credit-decision explanations.

e. Need of the study

This study is needed because digital lending is expanding faster than the mechanisms available to explain automated decisions to borrowers. The proposed work responds to four practical requirements in Indian financial services:

1. **Regulatory compliance and fair lending:** RBI expectations require lenders to give borrowers understandable reasons when credit is denied. Manual drafting of such explanations is slow and inconsistent, while unexplained automated rejection can weaken trust in digital lending. A controlled automated system is therefore required.
2. **Explainability-to-usability bridge:** SHAP and similar methods help analysts understand model behaviour, but their numerical outputs are not suitable for most borrowers. The study is needed to convert technical feature attributions into clear explanations that can be read and acted upon.
3. **Mitigation of algorithmic hallucination**: As financial institutions explore generative AI, there is a risk that an LLM may introduce unsupported rejection reasons or financial advice. This study is needed to test a design in which the language model is constrained to translate verified mathematical attributions rather than independently invent reasons for an adverse credit decision.
4. **Data privacy and local deployment:** Borrower data used in credit assessment is sensitive. Sending such data to external language-model services can create avoidable privacy exposure. A locally deployed explanation system can reduce this risk while supporting regulatory documentation.

f. Objectives

The main objective of this research is to build and test a privacy-preserving dual-agent system that operates on local infrastructure and converts credit-risk model attributions into borrower-facing explanations consistent with RBI fair-lending expectations.

To achieve this objective, the study will focus on the following tasks:

1. **Predictive underwriter component:** Train non-linear credit-risk models on alternative credit datasets and extract SHAP-based reasons for declined applications.
2. **Design of a zero-shot LLM Compliance Agent:** Develop a localized, instruction-tuned LLM component that interprets SHAP-based mathematical arrays and converts them into formal, actionable borrower-facing language aligned with RBI requirements.
3. **Privacy-preserving edge pipeline:** Design an API-independent workflow in which data remains inside the local computing environment during scoring, explanation, and notice generation.

4. **Hallucination-resistance validation:** Compare generated explanations with the original SHAP rankings using rank-correlation and language-similarity measures to assess whether the notice stays faithful to the model evidence.

g. Scope

The study is limited to explainability and regulatory communication for AI-enabled credit decisions in the Indian BFSI context. Its scope includes the following:

1. **Domain:** Alternative credit scoring used by Indian digital lenders and NBFCs.
2. **Technology:** SHAP-based explanation for non-linear credit models and local deployment of open-weight LLMs for explanation drafting.
3. **Regulation:** Borrower-facing explanation requirements associated with RBI fair-practice and digital-lending expectations.
4. **Evaluation:** Comparison of generated notices with SHAP rankings and reference notices using statistical and semantic measures.

h. Limitation of study

The following Limitations are specifically set in order to assure academic rigor and feasibility:

1. This project will not attempt to train a new foundational large language model from scratch but will instead utilize and prompt engineer existing open-weight models (e.g., Llama-3, Mistral, etc.).
2. The predictive ML model (Underwriter Agent) will be trained on public, anonymized alternative credit datasets (e.g., Kaggle Home Credit Default Risk) rather than private, live banking data, mimicking edge-case retail customers.
3. The research is purely on the translation/explainability of the credit decision, not on the optimization of the underlying predictive accuracy or the baseline default rate of the ML model itself.
4. The solution will be tested in a simulated, air-gapped edge computing environment to ensure data sovereignty and will not be implemented into a live, commercial banking production system.

i. Research questions

The project will try to solve the following key scientific questions:

RQ1: How to automatically translate post-hoc mathematical explainability vectors (SHAP) into human-readable RBI-compliant regulatory text without producing semantic or computational hallucinations?

RQ2: Can SHAP-guided financial narratives preserve mathematical fidelity and regulatory conformity when generated in English and Hindi borrower-facing outputs?

j. Research Hypotheses

Based on the highlighted gaps and research questions, the following hypotheses are developed:

H_0 (Null Hypothesis 1): The mathematical SHAP feature-importance rankings given by the ML model do not significantly correlate with the narrative reasoning generated by the localized LLM Compliance Agent.

H_1 (Alternate Hypothesis 1): An edge-deployed compliance agent constrained by SHAP inputs will exhibit a statistically significant positive rank with the underlying ML model, removing algorithmic hallucination in the generation of adverse action notices.

and

H_0 (Null Hypothesis 2): A localized multi-agent workflow independent of API cannot achieve a high degree of linguistic accuracy (ROUGE-L) of RBI-compliant regulatory text relative to cloud-based proprietary LLMs.

H_2 (Hypothesis 2): A multi-agent orchestration pipeline running on local hardware will produce RBI-compliant documentation with semantic accuracy (high ROUGE-L scores) comparable to baseline standards, thus meeting both regulatory transparency and stringent data sovereignty requirements.

k. Research Design

This research develops and validates the suggested multi-agent architecture using a quantitative, computational experimental method. The approach is targeted at the mathematical measurement of the fidelity of story translations with respect to baseline mathematical explainers.

1. **Research approach**: The study adopts an experimental computational methodology with two linked phases. In the deterministic phase, a predictive machine learning model generates credit decisions and SHAP-based feature attributions. In the generative phase, a locally deployed LLM converts the verified attribution rankings into borrower-facing regulatory narratives. To reduce hallucination risk, the LLM is restricted to zero-shot transformation of supplied reference rankings rather than independent risk assessment (Geng, Liu, Li, & Wang, 2026).

2. **Area of Study**: The area of study lies at the intersection of artificial intelligence—specifically agentic AI and natural language processing—and Explainable AI (XAI) as well as Regulatory Technology (RegTech), all applied in the context of the Indian Banking, Financial Services, and Insurance (BFSI) sector.

3. **Sample:** To comply with the rigorous data privacy rules in the BFSI sector, the study will use publicly available, highly dimensional alternative credit datasets that are suitable proxies for Indian FinTech lending data. The main dataset will be composed of datasets such as the Kaggle Home Credit Default Risk or the Lending Club Dataset. A statistically substantial random sample of at least 1,000 "rejected" loan applications will be drawn from this population to simulate the creation of Adverse Action Notices.

4. **Techniques**:

   The following computational approaches will be used in the research:

   - Ensemble Machine Learning: gradient boosting frameworks (XGBoost/LightGBM) to train the baseline predictive Underwriter Agent.
   - Post-Hoc Mathematical Explainability: The SHAP value attribution approach to compute the marginal contribution of each variable to the adverse credit decision.

- Prompt engineering and in-context learning: tightly constrained zero-shot prompts will guide the LLM Compliance Agent to convert mathematical vectors into RBI-aligned borrower-facing explanations.
- Multi-Agent Orchestration Programmatic chaining of outputs from the predictive model to the generative LLM pipeline

5. **Tools**:

   The following open-source scientific tools will be used for the computational experiments to guarantee reproducibility and data sovereignty:

   - Programming Environment: Python 3.x (Pandas, Scikit-learn, NumPy).
   - Explainability Libraries: The SHAP Python Library
   - Local LLM Hosting Run quantized, open-weight language models (e.g., Llama-3, Mistral) fully on local edge hardware with the Ollama framework, avoiding third-party API dependencies.
   - Agent Orchestration: Use of LangChain or AutoGen frameworks to orchestrate the interaction between underwriters and compliance agents.

6. **Measurement & Scaling**:

   For rigorous academic validation, the study is based on two main measurement scales:

   - Mathematical Concordance. Kendall's Rank Correlation Coefficient will be used to quantify the pairwise ordinal relationship between the ranked SHAP values and the feature-importance rankings obtained from the generated text of the LLM.
   - Accuracy: The closeness of the LLM's output with a "Golden Dataset" of humanly prepared, RBI-compliant notices will be evaluated using the ROUGE-L (Recall-Oriented Understudy for Gisting Evaluation) and Semantic Textual Similarity (STS) metrics.

7. **Data processing**:

   The raw credit datasets will be pre-processed using common data preprocessing techniques such as missing value handling, standardization of continuous variables, and one-hot encoding of categorical features. Once the ML model is trained, the data processing phase will separate the True Negatives and False Positives (rejected applications). For every application that is rejected, the system will loop through and

collect the top 3 negative SHAP vectors, converting from continuous numbers into the JSON/text array format required for ingestion into the LLM.

8. **Estimation and Analysis**:
   The analysis will computationally evaluate the outputs produced by the localized LLM. Natural language processing techniques will be used to extract the rejection grounds stated in each adverse action notice. These extracted reasons will then be ranked and compared with the original SHAP arrays. System performance will be assessed through the Hallucination Error Rate (HER), defined as the proportion of notices in which the LLM introduces a rejection reason that is absent from the SHAP baseline.

9. **Testing of Hypothesis**:
   The hypotheses developed will be tested on stringent statistical thresholds. We will test the null hypothesis (H0) that the LLM narrative does not correlate with the mathematical SHAP ranking using the estimated p-value associated with Kendall's Tau. If the test produces a high Kendall's Tau (> 0.90) with a p-value < 0.01, the null hypothesis will be rejected, mathematically establishing that the localized multi-agent framework successfully serves as an accurate, hallucination-free regulatory translator.

10. **Expected results**:
    The successful completion of this research is expected to lead to the following concrete and quantifiable outcomes:
    - **An Architected Edge-RegTech Framework**: The main outcome will be a fully functional, localized multi-agent pipeline (including an underwriter agent and a compliance agent) that is completely independent of external APIs, hence guaranteeing absolute data privacy for financial underwriting data.
    - **Mathematical Validation of Narrative Explainability**: This work will generate empirical evidence (using Kendall's Rank Correlation) quantifying the precise ability of instruction-tuned large language models to act as zero-shot narrative translators for mathematical SHAP values without the introduction of algorithmic hallucinations.
    - **Automated Regulatory Compliance Pipelines**: The study will produce a programmatic approach that can efficiently translate non-linear credit risk

attributions into the legal language necessary for the Reserve Bank of India's (RBI) Fair Practices Code, thus automating the drafting of Adverse Action Notices.

- **Cross-Lingual Implementation Evidence**: The research will yield comparative metrics on the effectiveness of generating SHAP-aligned regulatory alerts in English and Hindi, providing a framework for inclusive digital lending communication.

11. **Relevance of Anticipated Results**: The expected findings of the study are crucial for the academic community as well as the Indian Banking, Financial Services, and Insurance (BFSI) industry.
    - **Academic Relevance**: This research addresses a significant gap in the literature of Explainable AI (XAI). Although prior research has already mathematically proved the limitations of autonomous LLMs in the calculation of credit risk (Geng, Liu, Li, & Wang, 2026), this study contributes to the area by converting those results into a multi-agent system. It moves the discussion from mathematical transparency (e.g., raw SHAP values) to semantic explainability that is regulation-compliant.
    - **Industry and commercial relevance**: The findings may be useful for Indian fintech lenders and NBFCs that wish to use accurate non-linear models while meeting borrower-transparency expectations. By converting SHAP values into RBI-compliant language, the proposed framework can reduce the trade-off between predictive sophistication, regulatory accountability, and data-sovereign implementation.

12. **Time lines**: The proposed research will be performed in a typical 36-month doctorate schedule with well-defined milestones to achieve consistent progress towards the objectives.
    - Months 1 - 6 (Phase 1: Foundation): The student completes the needed courses, does a comprehensive literature review, and formalizes the evaluation measures.
    - Months 7 - 12 (Phase 2: Predictive Modelling) Data collecting, data preprocessing, and training of the baseline "Underwriter Agent" (XGBoost/LightGBM) extracting and normalizing SHAP values for the test set.
    - Months 13 - 18 (Phase 3: Agent Engineering) Ollama: Local deployment of the open weight LLM. Extensive prompt engineering and no parameter adjusting to operate as the "Compliance Agent."

- Months 19 - 24 (Phase 4: Orchestration & Edge Deployment) Development of the programmatic pipeline that connects the ML output to the LLM input without external APIs. Testing on air-gapped systems for data sovereignty verification.
- Months 25 - 30 (Phase 5: Evaluation & Analysis): Rigorous Mathematical Evaluation using Kendall's Rank Correlation and ROUGE-L. Performing cross-lingual (Hindi) translation tests.
- Months 31 - 36 (Phase 6: Documentation) Completion of the research thesis; publication of empirical results in peer-reviewed journals; preparation of the final defense.

13. **Tentative Chapter Outline**: The final thesis shall be organized to clearly demarcate the theoretical foundation, architectural engineering, and mathematical evaluation of the research.
    - Chapter 1: Introduction This chapter will explain the context of the Indian digital lending ecosystem, clarify the algorithmic transparency mandates of the RBI, formalize the issue statement, and outline the scope and objectives of the research.
    - Chapter 2: Literature Review A broad critical survey of previous works related to post hoc Explainable AI (XAI), functional limits of autonomous LLMs in financial applications, and privacy-preserving federated architectures.
    - Chapter 3: Proposed System Architecture and Methodology This chapter describes the overall research concept, mathematical models for assessment (Kendall's Tau, ROUGE-L), dataset parameters, and the plan for the API-independent, localized multi-agent orchestration pipeline.
    - Chapter 4: Predictive Underwriter Agent Implementation This chapter involves only the deterministic part of the design. It will describe the training of the non-linear machine learning classifiers (e.g., XGBoost) on alternative credit datasets and the systematic extraction of precise SHAP value arrays.
    - Chapter 5: Implementation of the LLM Compliance Agent Focus on the Generative Phase It will discuss quantized open-weight LLM deployment on edge hardware, limitations on rapid engineering, and zero-shot semantic translation of SHAP vectors into legal regulatory vernacular.
    - Chapter 6: Results and Discussions The empirical heart of the dissertation. This chapter will present the numerical evidence for the framework, including proof of

its resistance to hallucinations, accuracy scores for language processing, its effectiveness in Hindi, and the time required for it to function.

- Chapter 7: Conclusion and Future Scope A synthesis of how the research objectives were achieved, the academic and commercial implications for RegTech, constraints identified during the study, and particular avenues for future scholarly examination.

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